TSM CERAMIC

THE FUTURE
heat protection

EC Declaration of conformity

Description

Manufacturer	"INFO VIT" LTD Ukraine, Kharkov, 61058 , St. Kultura 9. apt. 1 phone.+38 (057) 705-04-85, Fax.+38 (057) 705-01-39
Authorized representative in EU	IK ICQC (International Center for Quality Certification) Latvia, LV2016, Jurmala, Skolas 63-19, str, LV40002155625, phone: +371 27168371, e-mail: eastern.branch@icqc.co.uk
Technical documentation has been compiled and is held by	IK ICQC (International Center for Quality Certification)
Product name	**TSM CERAMIC**
Brand	**"TSM CERAMIC Thermal insulation"**
Test report	№ M-4072/2010 19/10/10 (ÉMI ÉPÍTÉSÜGYI MINŐSÉGELLENŐRZŐ INNOVÁCIÓS NONPROFIT Kft)
The adhesive on the concrete hardness, MPa	≥0,3
The adhesive on the hardness of steel, MPa	≥0,3
Water permeability, kg.m -2 h -0,5	W2
Water vapor permeability	II
The adhesive on the concrete hardness with freeze-thawing cycle, in accordance with EN 13687-3	≥0,3
The coefficient of thermal conductivity W m -1 K -1	≤ 0,00177 W/mK
The fire class	E
Test report	№TMK/427/2010, 2010/03/19, Debreceni Egyetem Muszaki Kar, MSZ EN 1934, The coefficient of thermal conductivity ≤0,00177W/mK
The CE mark was first applied in:	2013

Conformity harmonized standards

EN 15824:2009, EN 13501-1:2007

Used other standards

MSZ EN 1542:2000, MSZ EN ISO 4624:2003, MSZ EN 1062-3:2009, MSZ EN ISO 7783-2:2000, MSZ EN 1542:2000, MSZ EN 12667:2001.

CE

We affirm that above mentioned product „TSM CERAMIC" manufactured by "INFO VIT" LTD fulfils the requirements of the Council Directive 89/106/EEC of 21 December 1988 on the approximation of laws, regulations and administrative provisions of the Member States relating to construction products

Vladimir Ginglatt, Managing Director

April 4, 2013 , Latvia, Jurmala

№JIC02250

The TSM Ceramic insulation material

Key Features

20 years factory Warranty

30-35 years Life expectancy

$\lambda = 0{,}001$ W / mK

Heat reflection > 80%

TSM CERAMIC
FOR THE WORLD

APPLICATION PRACTICE OF INNOVATIVE ENERGY-SAVING MATERIAL INNOVATIONS FOR PEOPLE

Liquid Ceramic Heat Insulation Coverage
TSM CERAMIC

Liquid ceramic heat insulation coverages (LCHI), including TSM Ceramic, appeared on Russian market at the beginning of 2000 years and their application on the objects of industry and buildings construction is swiftly broadening. What are these ultra-thin heat insulations are about?

Composition of TSM CERAMIC

TSM Ceramic consists of membrane forming material, ceramic filler and other special purpose additions - bioprotective, antifoam and antifreeze agents.

Styrene - acrylic latex is used in material as membrane forming component. Styrene (about 20%) gives hardness to material, and acrylic constituent (80%) - elasticity and weather resistance. Membrane has all properties of acrylic binder, such as weather resistance (guarantee on coverage no less than 10 years), elasticity bend (1 mm), wonderful adhesion, with hardness about 0,4 H/m2 at the same time, unlike acrylates (hardness 0,2 H/m^2) and no of enamels (0,1-0,3 H/m^2).

Ceramic filler. It is microspheres, filled by discharged gas, most of it is in ready made coverage (75% volume of dried-up material). It is this component, that gives high heat insulating efficiency to material.

Material also contains bioprotective additions (biocide, preservative, antifungal and others). These additions prevent the processes of fermentation, prolong period of material shelf storage, and also prevent development of fungus on a surface.

Water-based material is ecologically pure and does not contaminate the environment.

Principle of action of TSM Ceramic

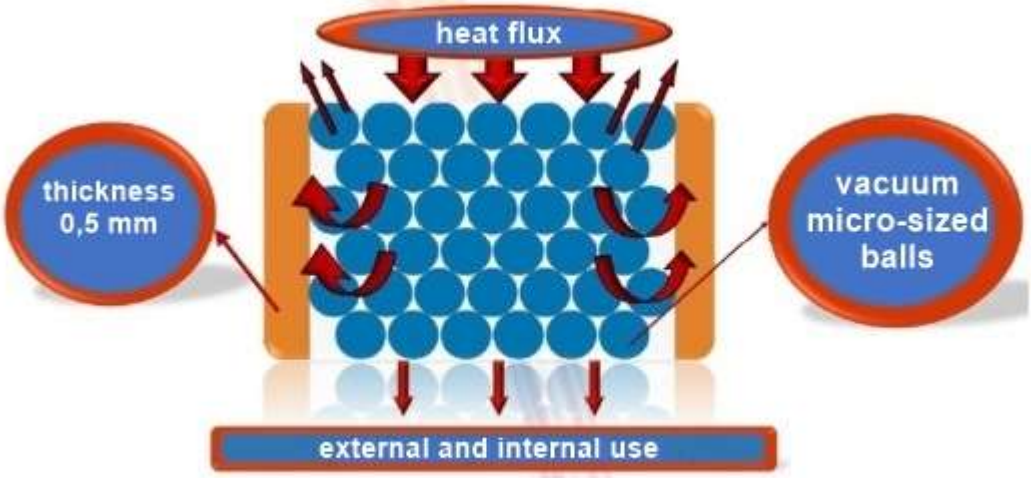
Thermo physical properties of TSM Ceramic substantially differ from similar properties of traditional heat insulators.

As it is known, the process of heat transfer in nature is fulfilled by plenty of physical phenomena:

- by heat conductivity of material itself;
- by convective heat exchange;
- by radiation.

Therefore resulting heat conductivity of any physical body is determined as a sum of these three constituents.

TSM Ceramic is a capillary and porous body. Its interporous space is filled by ceramic spheres that are being in discharged state. The gas, contained in microspheres, practically, does not conduct warm. In the process of heat transfer the wall of microsphere is heated at first, then gas inside of microsphere, then opposite wall, further, the heat is passed to the nearby microspheres with a loss of energy in every case. Taking into account that the diameter of sphere is microscopic (~10 mkm), a thermal stream has to do a long and winding way, although the thickness of material is only 0,5-3 mm. Thus, the rarefaction of interporous space significantly reduces the convective part of heat transition. It is known that at zero gravity convective heat exchange is absent at all.



TSM Ceramic, except heat transfer by heat conductivity, is semilucent for infrared (IR) radiation, i.e. has absorbing, radiating and dispersive properties, that fundamentally change the structure of thermal losses from a surface coverages, that consist of convective losses due to ablution of surface with surrounding air and radiation losses due to «relighting» of wall with terrene and welkin. For these reasons the heat emission coefficient of TSM Ceramic (1,29-2,2 W/^{of} m^2 ºC) is much below the level of other building and heat insulation materials (9-23 W/^{of} m^2 ºC)'

TSM Ceramic along with other hi-tech advantages possesses values, that practically do not meet in one material:

- heat insulation;
- waterproof;
- anticorrosive;

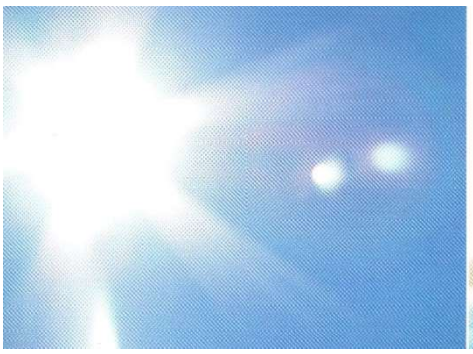
Research of liquid ceramic heat insulation (LCH) efficiency has some difficulties. Ultrathin heat- insulation materials has not entered in operating CHnn&i and GOST, there were no methodologies of research for similar materials, both as suitable measuring equipment.

Optical descriptions of TSM Ceramic are coefficients of absorption, reflection, radiation, that substantially differ from analogical descriptions for black metals, building materials or standard heat isolations. Therefore sensors of thermal streams that are used in practice of technical control should be applied with special calibration, because of high reflectivity and low level of TSM Ceramic coverage blackness, that result in substantial errors while using sensors, calibration of which was carried out with use of model blackbody emitters only for the values of radiation ability, that are close to 0.99, and are working in the spectral range of 7-18 mkm. Taking temperature probes of surface without introduction of special methodical amendments to the results of measuring, both as use of special filters to optical part of pyrometer, can result in substantial errors during fixing the final description.

At the same time, researches of TSM Ceramic, conducted in Moscow, Saint Petersburg, Kyiv, Minsk, Astana, Samara, Krasnoyarsk and other cities of Russia (based on energy saving at maintenance of programmed temperatures after putting TSM Ceramic of different thickness on enclosing structures, indicated high efficiency of material. According to these researches material thickness of 0,25 mm gives 15% of heat economy , 0,5 mm - 25%, 1 mm - 40%.

Technical specifications of TSM Ceramic

Name	Unit of measure	Value	Notes
Heat conductivity at 20°C, not more	W/ m °C	0,001	GOST 7076-87 (GOST-state standard)
Density in dry state	kg/m³	380-410	GOST 17177-94
Density in liquid state	kg/m³	470-590	GOST 17177-94
Coefficient of vapour permeability	mg/ m h Pa	0,0014	GOST 25989-83
Specific heat capacity	kJ/kg °C	1,08	
Heat-resistance at temperature of 260°C	Absence of cracks, swelling and stratifications		
Water absorption	g/sm³	0,03	GOST 11529-86
Relative elongation at break, not more	%	>12	GOST 11262-80
Relative elongation at break after accelerated aging (10 years), not more	%	>12	
Linear elongation	%	65	GOST 11262-80
Coupling durability at tearing, no less			
- with metal	Mpa	1,53	GOST 15140-78
- with concrete		1,84	
- with wood		1,84	
Tension durability, no less			
- after infliction	Mpa	2,0	GOST 11262-80
- after accelerated aging (10 years)		3,0	
Shock durability	Kg/cm	50	GOST 4765-73
Whiteness % of diffusive reflection			
- after infliction	%	93,0	GOST 896-69
- after accelerated aging (10 years)		90,0	
Temperature of transportation and storage	C°	+ 5	
Temperature of surface at material infliction	C°	from + 1 to +150	
Temperature of exploitation		from – 60 – to 250	



Energetic

Calculation formulas for flat wall are much more simple comparing with formulas for cylindrical object. Usually formulas of flat wall can be applied if diameter of the insulated wall is more than 2000 mm. Thickness TSM Ceramic isolation changes within the limits of 0,5-3,5 mm range. Consequently when applying TSM Ceramic it would be reasonable to use the formulas of calculations for flat walls on cylindrical objects.

Thus, the calculation of thickness of TSM Ceramic heat-insulation coverage on hot surfaces should be performed according to Cn 41-103-2000 using next formula:

$$\delta_{TSM} = \lambda_{TSM} (T_H - T_0) / \alpha_{TSM(1,2)}(T_n - T_o$$

$$Q = \alpha_{TSM(1,2)}(T_n - T_o) \text{ or}$$

$$Q = (T_n - T_o) / (1/\alpha_B + 1/\alpha_{TSM(1,2)} + \delta_{Tp} / \lambda_{Tp}$$

where

δ - is thickness of isolation TSM Ceramic (mm);

λ_{TSM}=0,001 - coefficient of heat conductivity for TSM Ceramic (W/sq.m. °C)

α_{TSM1} 1,29 - coefficient of heat conductivity for TSM Ceramic at indoor material infliction (W/sq.m. °C)

α_{TSM2} 2,2 - coefficient of heat conductivity for TSM Ceramic at outdoor material infliction (W/sq.m.°C)

T_n Temperature of carrier (°C)

T_n Temperature of surface after infliction of TSM Ceramic according to sanitary and hygienic requirements (°C)

T_o Temperature of environment (°C)

Q Heat losses on 1 sq.m. of surface isolated by TSM Ceramic (W);

During calculation of coverage thickness on objects located indoors, ambient temperature should be +18°C, +20°C.

During calculation of coverage thickness on objects located outdoors, value of ambient temperature should equal to average annual temperature in this Region.

Calculation methods of TSM Ceramic coverage thickness for cold surfaces (from condensate and formation of ice).

As practice indicates, the higher humidity of air in apartment, the thicker isolation should be. However there are such conditions, that make impossible the removal of condensate or ice from the surface of object. Such conditions arise when the gradient of temperatures is more than 35°C and air humidity is more than 70%.

Calculations of TSM Ceramic isolation thickness should be conducted according to CHnn 41-03¬2003 year, using next formula:

$$\delta_{TSM} = \lambda_{TSM} / \alpha_{TSM} \{((T_o - T_H)/(T_o + T_H) - 1\}$$

where:

δ is thickness of isolation TSM Ceramic (mm);

λ_{TSM} =0,001 - coefficient of heat conductivity for TSM Ceramic (W/sq.m. °C)

α_{TSM} 1,29 - coefficient of heat transfer for TSM Ceramic (W/sq.m. °C)

T_n Temperature of carrier (°C)

T_o Temperature of environment (°C)

(T_n-T) - calculated drop of temperatures on defined values (in %) relative humidity of surrounding air (°C);

Estimated period $(T_n - T)$ is defined according to table nr 1

T_o, ^{0}C	Relative air humidity					
	40	50	60	70	80	90
10	13,4	10,4	7,8	5,5	3,5	1,6
15	14,2	10,9	9,1	5,7	3,6	1,7
20	14,8	11,3	8,4	5,9	3,7	1,8
25	15,3	11,7	8,7	6,1	3,8	1,9
30	15,9	12,2	9	6,3	4	2

In general calculations on isolation thickness are done according to Cn 41-103-2000 using next formula:

$$\delta=\lambda/\alpha\ ((T_n\ T_H)/(T_n\ T))$$

Heat-insulation of pipelines and equipment

Removal of condensate

Thermal insulation of pipelines in households

Petroleum industry

TSM Ceramic can successfully be used for heat insulation of petroleum and gas pipelines of both surface and underground gasket. Thus cost of executed works with use of TSM Ceramic material on insulated object is on 30% cheaper comparing with rockwool and polyurethane foam. The second economy is diminishing of energy losses during exploitation of objects - to 40%. At the present moment TSM Ceramic positively proved itself positively on objects «Gazprom», «Rosneft», «Lukoil».

Heat insulation of capacities for water and oil products

Engineering

TSM Ceramic is used for warming of carriages, diesel engines.

Warming of carriage cabins

Shipbuilding

In shipbuilding material TSM Ceramic is used for warming of cabins, interior parts of corridors on ships, waterproofing.

Heat insulation of ships and interior parts of corridors on ships

Example of joint use of polyurethane foam and liquid ceramic coverage TSM Ceramic in heat insulation of pipelines

We will take a specific example:

Outdoor heating pipeline D=820 mm.

T_H=+150°C is a temperature of carrier on exit from boiler room;

T_O=+25°C is a temperature of surrounding air. We take a summer period for calculation, because at this time of year the surface of pipe will has the highest temperatures;

T_n- temperature of pipe surface after infliction of TSM Ceramic;

λ_{TSM} = 0,001 - coefficient of TSM Ceramic heat conductivity (W/m°C);

α_{TSM} = 2,25 - coefficient of TSM Ceramic heat emission (W/m°C);

TSM Ceramic is inflicted on a pipe both as anticorrosion and heat insulation material. In our example it should reduce the temperature of pipe surface to the values, that are acceptable for subsequent use of polyurethane foam.

Taking into consideration that the surface of pipe can have roughness up to 0,15 mm, we will inflict 0.6mm of TSM Ceramic material for high-quality coverage of the pipe.

The thickness of TSM Ceramic coveraee is calculated with a formula:

$$\delta_{TSM}=\lambda_{TSM}(T_H-T_n)/\alpha_{TSM}(T_n-T_o)$$

As we can see, at the above mentioned conditions, the unknown value is a temperature of pipe surface) after infliction of NEW TSM Ceramic δ_{TSM}=0,6 mm thick.

Consequently:

$$0{,}0006=0{,}001*(150-T_n)/2{,}5*(T_n-25)=0{,}15-0{,}001*T_n/2{,}5T_n-32{,}25;$$

$$0{,}0025*T_n=0{,}1875;$$

$$T_n=75^oC$$

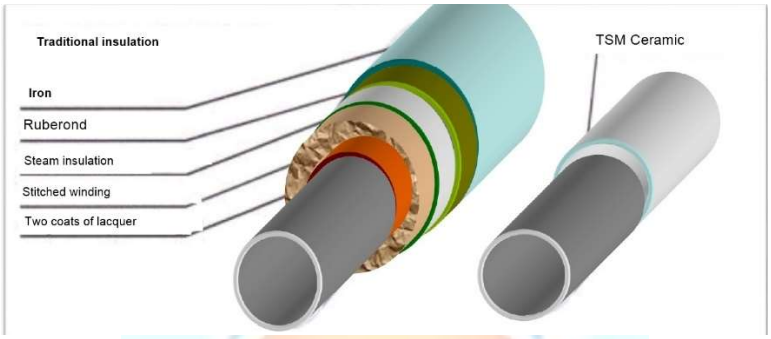
The temperature of surface +75 °C is fully acceptable for subsequent additional heat insulation with polyurethane foam.

Heat insulation of steam lines and pipelines

Cost saving

1. Reduction of labor and time capacity of TSM Ceramic use, due to easiness and simplicity of work with material.
2. Cost saving on expenses for repair of pipeline after expiration of warranty period, due to absence of necessity to uninstall the old insulation and performing works on preparation of old pipeline for insulation.
3. Cost reduction on economy of thermal energy in pipelines, steam-boilers and so on, due to high heat-insulation descriptions of TSM Ceramic and complete insulation of pipelines, steam- boilers, bolts, transitions and so on, even in places with the most difficult access.

4. Possibility to inflict TSM Ceramic directly on hot surface, without shutting down the heating system network or steam-boiler.
5. Cost reduction on installation of heat-insulation, due to diminishing of technological operations, related to warming of pipelines and so on, when using TSM Ceramic as an insulation.
6. Cost reduction on repair of pipeline in case of emergency situations, due to reduction of time for search of flow, fistula and no need to uninstall the old isolation.
7. Cost reduction on repair of heat insulation, due to the increase of warranty period comparing with standard insulations.
8. No expenses on restoration of insulation because of no possibility for its secondary use.



Building industry.

TSM Ceramic is used in building not only as heat-insulating coverage, but also as a waterproof material. Presence of latex in material provides it with low water absorption capacity.

Possibility to use TSM Ceramic as protection from condensate formation in apartments, enables antifreeze protection, both as removal of mycotic formations and mould.

= 0.002- coefficient of TSM Ceramic heat conductivity (W/sq.m °C) b = 8.7 - coefficient of heat transfer of fencing construction outdoor surface (W/sq.m °C).

Heat insulation of frontage and painting with any color

Heat insulation and water protection of roof

Heat insulation of walls, ceilings and floors

Warming of the balconies and elimination of cold bridges

Thermal insulation of cottages

Vapour permeability of TSM Ceramic

TSM Ceramic has low coefficient of vapour permeability. Therefore we will consider whether it will not be hindering the diffusion of moisture vaporization from the layer of wall to outside, as long as resistance of vapour permeability should be diminished because of internal layer of multilayer construction to outer one.

Advantages of TSM Ceramic comparing with standard insulation

1. Highly resistant to weather conditions and drops of temperatures.
2. Highly resistant to influence of sun radiation.
3. Extremely low coefficient of heat conductivity.
4. Durable - with warranty term of 10 years, term of exploitation at outdoor use - over 20 years.
5. High degree of adhesion.
6. Corrosion resistant, waterproof.
7. High temperature of exploitation - up to +260 °C.
8. Works on installation of heat insulation are not laborious.
9. Easiness of repair activities and finding out the flows.
10. Resistant to mechanical damages.
11. Possibility to use the insulation on pipelines and objects with complicated configuration, in places of difficult access .
12. Ecologically clean and fireproof material.
13. Cost and period of installation works are 30% lower comparing with traditional heat-insulation.

Most effective spheres of TSM Ceramic application:

- Walls of apartments and industrial buildings, both from internal and external sides.
- Roofs of apartments and industrial buildings, both from internal and external sides.
- Metallic constructions.
- Angaras and garages.
- Crane girders.
- Lower part of bridges (lowers the freezing).
- Pipelines of thermal heating systems.
- Steam and gas pipelines.
- Systems of air conditioning.
- Pipes with cold water (for prevention of condensation).
- Hydrants, water heaters and boilers.
- Heat exchanger
- Steam boilers.
- Underground and ground oil pipelines.

- Hot chemical mixer tanks.
- Capacities and tanks for storage of water, chemical reagents etc.
- Cooling chambers.
- Coverage of internal side of motor compartment corps, roofs of transport vehicles.
- Coverage of internal side of military and special purpose facilities.
- Refrigerators.
- Motor car and railway cisterns for different liquids.
- Engine-rooms of ships.
- Decks and internal part of ship corps.